

Spectral Costs of Multinomial and Residual Resampling in High-Dimensional Particle Ensembles: A Free-Probability Analysis

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Abstract—Resampling is the standard remedy against weight degeneracy in particle filters, yet its effect on the spectrum of the particle second-moment matrix is poorly understood. We study a single resampling step for a d -dimensional ensemble of N particles with $d/N \rightarrow c$. For any unbiased scheme, the resampled empirical second-moment matrix dominates its pre-resampling counterpart in conditional convex order, while the conditional mean is preserved; the excess second spectral moment equals the Frobenius energy of the resampling perturbation. For multinomial resampling, the empirical count law converges to the mixed-Poisson law determined by the limiting weight law. For residual resampling, the count law is determined by the joint limit of floor–fraction pairs: we classify all residual limits of weight sequences converging to δ_1 in law as a one-parameter family between the deterministic identity and the full multinomial count law. Each count law feeds a common covariance-side spectral map, yielding exact consequences: a closed-form zero-eigenvalue mass, a rank-and-spectral-gap transition with a universal lower-edge benchmark, computable regular edges, unbounded right support, and scheme-specific second-moment costs. The effective sample size determines one bulk statistic but cannot determine the limiting spectrum. Pre-registered numerical experiments confirm the theory at every level.

Index Terms—Particle filtering, resampling, random matrix theory, free probability, limiting spectral distribution, effective sample size.

I. INTRODUCTION

PARTICLE filters approximate posterior distributions by weighted particles [1] [2]. Resampling, the redrawing of particles according to their weights, is the standard remedy against weight degeneracy [3]. Its effect on the geometry of the ensemble is considerably less well understood: specifically, its effect on the spectrum of the ensemble second-moment matrix, which governs the conditioning of subsequent computations, and on the rank of the ensemble subspace. This paper isolates a single resampling step and determines how resampling transforms the limiting spectral distribution of the empirical second-moment matrix in a high-dimensional particle ensemble.

A. Finite-sample results

For any unbiased scheme, the resampled matrix dominates the pre-resampling matrix in conditional convex order (Theorem 3). The principal finite-sample result is Proposition 4, which identifies the excess second spectral moment with the expected Frobenius energy of the resampling perturbation.

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The resulting expression is a quadratic form in the count covariance, and it separates the contribution of the particle geometry, carried by $G \circ G$, from that of the mechanism noise, carried by $\text{Cov}(\mathbf{n} \mid X_N, w)$.

B. High-dimensional limit

The covariance-side step is classical: weighted sample-covariance matrices converge to a deterministic law $\mathcal{T}_c(\nu)$ characterized by a Silverstein-type fixed-point equation, equivalently by a free-multiplicative-convolution relation (Definition 8). The contributions of this paper concern the count-law side.

For multinomial resampling the limiting count law is determined by the limiting weight law ρ alone (Theorem 7). For residual resampling it is not. The floor–fraction map is discontinuous at the integers, and the limiting count law is determined instead by the joint limit π of floor–fraction pairs (Theorem 9). The strongest form of this dependence is a complete classification (Theorem 10): the residual limits of weight sequences converging to δ_1 constitute the one-parameter family $(1-\alpha)\delta_1 + \alpha \text{Poi}(1)$ with $\alpha \in [0, 1]$, ranging from the deterministic identity at $\alpha = 0$ to the full multinomial count law at $\alpha = 1$. The endpoint $\alpha = 1$ is attained by a bounded two-level construction, which establishes that the phenomenon originates at the integer boundary rather than in heavy tails.

The resulting framework is

$$\mu_{s,H} = H \boxtimes \mathcal{T}_c(\nu_s), \quad (1)$$

with

$$\nu_{\text{pre}} = \rho, \quad \nu_{\text{multi}} = \text{MixPoi}(\rho), \quad \nu_{\text{res}} = \mathcal{L}_\pi(K + \text{Poi}(U)), \quad (2)$$

where the last of these depends on finer information than ρ alone.

C. Consequences

Composed with the covariance-side spectral map, these count laws yield a closed-form zero-eigenvalue mass; a rank-and-spectral-gap transition at $p_s = 1 - \nu_s(\{0\})$, with an explicit Marchenko–Pastur lower-edge benchmark [4] and computable regular edges; exact second-moment increments ca_1^2 for multinomial resampling and $ca_1^2 \mathbb{E}_\pi U$ for residual resampling; an asymptotic rank-preservation ordering $\mu_{\text{res}}(\{0\}) \leq \mu_{\text{multi}}(\{0\})$; and a finite-sample Frobenius ordering of the two schemes. For multinomial resampling the limiting right support is unbounded, so that the pseudo-condition number diverges in probability and bulk extent must be interpreted as

a quantile notion. Sections V–VII state these results precisely; residual limits may instead have bounded support.

The results order the spectral observables by the information each requires. The bulk second moment, and therefore the effective sample size (ESS), depends only on $r_2 := \int \lambda^2 \rho$. The zero mass and the rank threshold depend only on $\mathcal{L}_\rho(1)$. The spectral edges and the full limiting spectral distribution depend on the entire count law. The choice of mechanism is therefore not observable through ESS [5].

D. Relation to prior work

Convex-order comparisons of resampling schemes are due to Gerber, Chopin and Whiteley [6], and the inadequacy of ESS as a sole diagnostic was argued by Elvira, Martino and Robert [7]. Weighted and separable covariance models are classical [8]–[12], and deterministic resampling schemes were surveyed by Li, Sattar and Sun [13]. The bootstrap spectral analysis of El Karoui and Purdom [14] concerns the statistical bootstrap rather than the bootstrap particle filter [15], [16]. The elements introduced here are the count-law limits of resampling mechanisms, including the integer-boundary classification for residual resampling, together with the mechanism-level spectral consequences derived from them.

E. Notation

Eigenvalues are ordered $\lambda_1(M) \geq \dots \geq \lambda_d(M)$, and $\lambda_{\min}^+$ denotes the smallest positive eigenvalue; $\mathcal{S}_+^d$ denotes the positive semidefinite matrices and tr the trace. Further, δ_x is the Dirac mass, $\Rightarrow$ weak convergence, $\text{MP}(y)$ the Marchenko–Pastur law [4] with ratio y and mean 1, $\boxtimes$ free multiplicative convolution, D_c the push-forward under $x \mapsto cx$, $G = X^\top X$ the Gram matrix, $A \circ B$ the entrywise product, $\nu^{(M)} := (t \mapsto t \wedge M) \# \nu$, and $\kappa^+(S) := \lambda_{\max}(S) / \lambda_{\min}^+(S)$. We write $\text{frac}(x) := x - \lfloor x \rfloor \in [0, 1]$, $\mathcal{L}_\rho$ for the Laplace transform of ρ , and $\mathcal{L}(\cdot)$ for the law of a random variable. Finally, $c_N := d/N$, $r_2 := \int \lambda^2 \rho(d\lambda)$, and π denotes a probability law on $\mathbb{Z}_{\geq 0} \times [0, 1]$ whose second coordinate is permitted to attain 1, with $\mathbb{E}_\pi U := \int u \pi(dk, du)$.

II. ONE-STEP RESAMPLING MODEL, ASSUMPTIONS, AND CONVENTIONS

A. Particles

Let $X_N = [x_1, \dots, x_N] \in \mathbb{R}^{d \times N}$ with $X_N = \Sigma_N^{1/2} Z_N$, where $\Sigma_N \in \mathcal{S}_+^d$ is a deterministic population second-moment matrix with empirical spectral law H_N , and where Z_N has independent and identically distributed entries drawn from a fixed distribution with mean 0, variance 1, and the moment conditions stated with each result.

The matrix Σ_N represents the deterministic correlation structure, which in filtering terminology is the background or forecast covariance; its spectral law H_N is the free parameter of the asymptotic analysis. The matrix Z_N is a whitened fluctuation matrix, so that the columns $x_i = \Sigma_N^{1/2} z_i$ are independent, identically distributed and centered, with $\mathbb{E}[x_i x_i^\top] = \Sigma_N$. The separable covariance model assigns the population spectrum H and the count law ν to two separate

channels, which is what permits the free-probability analysis. No particle model is required for the finite-sample results of Section III; the model is invoked only where spectra are concerned.

B. Conditioning conventions

The finite-sample statements of Section III are conditional on (X_N, w) . Asymptotic statements are taken under the joint randomness of the particle matrix and the resampling mechanism.

C. Weights and exogeneity

Let $w = (w_1, \dots, w_N)$ with $w_i \geq 0$ and $\sum_i w_i = 1$, and set $\lambda_i := N w_i$ and $\rho_N := \frac{1}{N} \sum_i \delta_{\lambda_i}$. Throughout, exogeneity is understood as follows. For the asymptotic analysis the weights are either deterministic, which is the convention adopted in the main text, or random and independent of Z_N , in which case all weight-law convergences hold in probability and the limit laws are deterministic. Multinomial and residual resampling employ auxiliary randomness independent of Z_N , so that the count vector is independent of the particle matrix conditionally on the weights. No such independence is required for the finite-sample results.

D. Counts

A resampling scheme produces $\mathbf{n} = (n_1, \dots, n_N) \in \mathbb{Z}_{\geq 0}^N$ satisfying

- (A1) $\sum_i n_i = N$ almost surely, and
- (A2) $\mathbb{E}[n_i | X_N, w] = N w_i$.

E. Second-moment matrices

The weighted (pre-resampling) and resampled (post-resampling) empirical second-moment matrices are

$$S_{w,N} = \sum_i w_i x_i x_i^\top = \frac{1}{N} X_N \Lambda_N X_N^\top,$$

$$S_{r,N} = \frac{1}{N} \sum_i n_i x_i x_i^\top = \frac{1}{N} X_N D_N X_N^\top,$$

with $\Lambda_N = \text{diag}(\lambda_i)$ and $D_N = \text{diag}(n_i)$. Both matrices are uncentered. Centering perturbs each matrix by a term of rank at most one and therefore leaves the limiting spectral distribution unchanged as $d \rightarrow \infty$ whenever that distribution exists. The finite-sample conditional inequalities are, however, sensitive to centering, since $\mathbb{E}[S_{r,\text{centered}} | X_N, w] = S_{w,\text{centered}} - \text{Cov}(\bar{x}_r | X_N, w)$.

F. Asymptotic assumptions

- (A3) $d/N \rightarrow c \in (0, \infty)$.
- (A4) For multinomial resampling, $\rho_N \Rightarrow \rho$ with $\int \lambda^2 \rho_N \rightarrow \int \lambda^2 \rho < \infty$.
- (A4R) For residual resampling, $\pi_N := \frac{1}{N} \sum_i \delta_{(\lfloor \lambda_i \rfloor, \text{frac}(\lambda_i))} \Rightarrow \pi$ on $\mathbb{Z}_{\geq 0} \times [0, 1]$, with $\int \lambda^2 \rho_N \rightarrow \int \lambda^2 \rho < \infty$, where ρ is the push-forward of π under $(k, u) \mapsto k + u$.
- (A5) $H_N \Rightarrow H$ with H compactly supported and $\sup_N \|\Sigma_N\|_{\text{op}} < \infty$; we set $a_1 := \int \tau H(d\tau)$ and $m_{2,H} := \int \tau^2 H(d\tau)$.

We record two consequences of these assumptions. The second-moment assumption implies $\max_i w_i \leq (\sum_i \lambda_i^2)^{1/2}/N = O(N^{-1/2}) \rightarrow 0$, so that diffuseness of the weights is automatic and need not be imposed separately. Finiteness of the limiting first moments follows by uniform integrability, giving $\int \lambda \rho(d\lambda) = 1$ and $\mathbb{E}_\pi(K + U) = 1$. Assumption (A4) does not imply (A4R), because the floor-fraction map is discontinuous at the positive integers; see Theorem 10 and Section VII.

G. Moment requirements

The conditional identities of Section III require no particle model. Convergence of the limiting spectral distribution requires independent, identically distributed entries with finite fourth moment. Convergence of the empirical second spectral moments requires, in addition, the count-tail control established in the supplementary material, or else Gaussianity. Extreme-eigenvalue statements invoke the extreme-spectrum theorem with entries from a fixed distribution.

III. FINITE-SAMPLE SPECTRAL INEQUALITIES

All statements in this section are conditional on (X_N, w) , hold for every d and N , and apply to every scheme satisfying (A1)–(A2).

Lemma 1 (Conditional unbiasedness identity). $\mathbb{E}[S_{r,N} \mid X_N, w] = S_{w,N}$.

Definition 2 (Conditional convex order [17]–[19]). We write $S_{w,N} \preceq_{\text{cx}} S_{r,N}$ conditionally on (X_N, w) if $F(S_{w,N}) \leq \mathbb{E}[F(S_{r,N}) \mid X_N, w]$ almost surely for every convex $F : S_+^d \rightarrow \mathbb{R}$ for which the required expectations exist. Convexity is understood in the ordinary affine sense on the positive semidefinite cone; no operator convexity or Löwner-order comparison is asserted.

Theorem 3 (Scheme-agnostic Jensen principle [20]). Under (A1)–(A2), $S_{w,N} \preceq_{\text{cx}} S_{r,N}$ conditionally on (X_N, w) .

Theorem 3 applies in particular to $F(M) = \text{tr } \varphi(M)$ for scalar convex φ , by Davis trace convexity [21]. Admissible choices include $\varphi = x^q$ for all $q \geq 1$ and $\varphi = x \log x$, together with $F = \lambda_{\max}$ and $F(M) = -\log \det(M + \varepsilon I)$ for $\varepsilon > 0$; the last of these states that the log-determinant decreases in conditional expectation. We note that x^q with $q > 2$ is not operator convex on $(0, \infty)$, whereas trace convexity remains available.

Proposition 4 (Exact second spectral moment cost [22], [23]). With $G = X_N^\top X_N$, for any scheme satisfying (A1)–(A2),

$$\begin{aligned} \mathbb{E}[\text{tr } S_{r,N}^2 - \text{tr } S_{w,N}^2 \mid X_N, w] \\ = \mathbb{E}[\|S_{r,N} - S_{w,N}\|_F^2 \mid X_N, w] \\ = \frac{1}{N^2} \text{tr}[(G \circ G) \text{Cov}(\mathbf{n} \mid X_N, w)] \geq 0. \end{aligned} \quad (3)$$

The second expression admits an error-energy interpretation and renders the nonnegativity immediate. For multinomial resampling,

$$\begin{aligned} \mathbb{E}[\text{tr } S_{r,N}^2 - \text{tr } S_{w,N}^2 \mid X_N, w] \\ = \frac{1}{N} \sum_i w_i \|x_i x_i^\top - S_{w,N}\|_F^2 \\ = \frac{1}{N} \left[\sum_i w_i \|x_i\|^4 - \text{tr } S_{w,N}^2 \right] \\ \geq \frac{1}{N} \text{Var}_w(\|x_i\|^2) \geq 0. \end{aligned} \quad (4)$$

Remark 5 (Vanishing of the cost). The cost is zero precisely when resampling leaves the second-moment matrix unchanged almost surely. Two distinct mechanisms produce this situation: deterministic counts, as arise for residual resampling when every $\lambda_i \in \mathbb{Z}_{\geq 0}$, or for stratified, systematic and residual resampling under uniform weights; and degenerate particle configurations. For multinomial resampling, the geometric form (4) vanishes if and only if $x_i x_i^\top = S_{w,N}$ for every i with $w_i > 0$, a condition that covers the all-zero and the $\pm v$ collinear configurations.

Proposition 6 (Population-averaged cost and diffuse-weight limit). Under multinomial resampling, independent and identically distributed entries with kurtosis $\kappa_4 := \mathbb{E}Z_{11}^4 - 3$, and weights independent of the particle matrix, we have exactly, with $s_{2,N} := \sum_i w_i^2$,

$$\begin{aligned} \frac{1}{d} \mathbb{E}[\text{tr } S_{r,N}^2 - \text{tr } S_{w,N}^2 \mid w] \\ = \frac{1-s_{2,N}}{Nd} \left[(\text{tr } \Sigma_N)^2 + \text{tr } \Sigma_N^2 + \kappa_4 \sum_{j=1}^d (\Sigma_N)_{jj}^2 \right]. \end{aligned} \quad (5)$$

Under (A3) and (A5) the bracketed terms satisfy $\text{tr } \Sigma_N^2 = O(d)$ and $\sum_j (\Sigma_N)_{jj}^2 = O(d)$, whence

$$\begin{aligned} \frac{1}{d} \mathbb{E}[\text{cost} \mid w] &= c_N \left(\frac{\text{tr } \Sigma_N}{d} \right)^2 (1 - s_{2,N}) + O(N^{-1}) \\ &= c a_1^2 (1 - s_{2,N}) + o(1), \end{aligned} \quad (6)$$

which reduces to $c a_1^2$ in the diffuse regime. The statement is specific to multinomial resampling; residual resampling under exactly uniform weights, for instance, leaves the matrix unchanged.

IV. COUNT LAWS AND THE SPECTRAL MAP

For a probability law ρ on $[0, \infty)$ with mean 1, the mixed-Poisson law is defined by

$$\text{MixPoi}(\rho) : \quad \nu(\{k\}) = \int_0^\infty e^{-\lambda} \frac{\lambda^k}{k!} \rho(d\lambda), \quad k \in \mathbb{Z}_{\geq 0}, \quad (7)$$

and has mean 1, zero mass $\mathcal{L}_\rho(1)$, and second moment $b_2 := \int k^2 \nu(dk) = 1 + r_2$.

Theorem 7 (Multinomial count law). Under (A4), for multinomial resampling, in probability,

$$\nu_N := \frac{1}{N} \sum_i \delta_{n_i} \Rightarrow \nu = \text{MixPoi}(\rho), \quad \frac{1}{N} \sum_i n_i^2 \rightarrow b_2. \quad (8)$$

Exactly uniform weights, and more generally any weight sequence satisfying (A4) with $\rho = \delta_1$, yield $\nu = \text{Poi}(1)$, so that a fraction e^{-1} of the particles is not selected.

The transfer of the classical theorem to a random and unbounded count diagonal proceeds by a Poissonization coupling together with truncation and rank control; the coupling controls weak spectral convergence but does not by itself control second spectral moments. Both steps are stated and proved in the supplementary material.

Definition 8 (Covariance-side spectral map [24]). *For a probability law η on $[0, \infty)$, define $\mathcal{T}_c(\eta)$ as the probability law determined by*

$$c\mathcal{T}_c(\eta) + (1-c)\delta_0 = D_c(\eta \boxtimes \text{MP}(1/c)), \quad (9)$$

or equivalently by the fixed-point equation

$$z = -\frac{1}{m(z)} + \int \frac{t}{1+ctm(z)} \eta(dt),$$

$$m(z) = \int \frac{\mathcal{T}_c(\eta)(dx)}{x-z}, \quad z \in \mathbb{C}^+, \quad (10)$$

taken with the branch of a probability-measure Stieltjes transform, namely $m : \mathbb{C}^+ \rightarrow \mathbb{C}^+$ with $\lim_{v \rightarrow \infty} ivm(iv) = -1$. Mean one is not required of η ; the truncated count laws $\nu^{(M)}$ have mean at most one and are never rescaled.

Three properties of Definition 8 are used repeatedly. First, (9) is an identity of measures rather than a convex-mixture representation when $c > 1$; equivalently $\mathcal{T}_c(\eta) = \frac{1}{c}D_c(\eta \boxtimes \text{MP}(1/c)) + (1-\frac{1}{c})\delta_0$ as measures, the negative coefficient for $c < 1$ being absorbed by the zero atom of the product law, in accordance with Proposition 14. Second, $\mathcal{T}_c$ is injective on all probability laws on $[0, \infty)$. Third, for numerical evaluation the companion-side transform is best computed from m through the auxiliary equation $z = -c/u + \int t(1+tu)^{-1}\eta(dt)$ with $u = cm$, followed by $e = u + (c-1)/z$. The classical companion fixed-point equation governs a different object, namely the companion transform of the W -side model, and several mutually incompatible forms coincide at $\eta = \delta_1$. Recovery of the Marchenko–Pastur law alone therefore does not validate a solver.

Theorem 9 (Residual count law). *Under (A4R), for residual resampling with $n_i = \lfloor \lambda_i \rfloor + \tilde{n}_i$, where $\tilde{\mathbf{n}} \sim \text{Multinomial}(R_N; u/R_N)$ with $u_i := \text{frac}(\lambda_i)$ and $R_N := \sum_i u_i$ when $R_N > 0$, and $\tilde{\mathbf{n}} := 0$ when $R_N = 0$, we have in probability*

$$\nu_N \Rightarrow \nu_{\text{res}} := \mathcal{L}(K + P_U), \quad (K, U) \sim \pi, \quad (11)$$

where $P_U \mid (K, U) \sim \text{Poi}(U)$ is a conditional Poisson variable rather than an independent one of fixed parameter, and $\frac{1}{N} \sum_i n_i^2 \rightarrow b_2(\nu_{\text{res}}) = r_2 + \mathbb{E}_\pi U$. All regimes of R_N are covered, namely $R_N = 0$, $R_N \rightarrow \infty$ with $R_N = o(N)$, and $R_N \asymp N$.

Theorem 10 (Classification of residual limits under $\rho = \delta_1$). *If $\rho_N \Rightarrow \delta_1$ and (A4R) holds with limit π , then π is supported on $\{K + U = 1\}$, which within $\mathbb{Z}_{\geq 0} \times [0, 1]$ consists of the two points $(1, 0)$ and $(0, 1)$. Consequently*

$$\pi = (1-\alpha)\delta_{(1,0)} + \alpha\delta_{(0,1)},$$

$$\nu_{\text{res}} = (1-\alpha)\delta_1 + \alpha\text{Poi}(1), \quad \alpha \in [0, 1], \quad (12)$$

with $\mathbb{E}_\pi U = \alpha$, $p_{\text{res}} = 1 - \alpha/e$, and second-moment cost $ca_1^2\alpha$.

Every $\alpha \in [0, 1]$ is realizable by a sequence satisfying both (A4) and (A4R). The value $\alpha = 0$ is attained by exactly uniform weights. The value $\alpha = 1$ is attained by the bounded two-level endpoint construction $\lambda_i = 1 - \frac{1}{N}$ for $i < N$ and $\lambda_N = 2 - \frac{1}{N}$, which has mean exactly one and satisfies $\max_i \lambda_i < 2$; the phenomenon is therefore driven by the integer boundary and not by heavy tails. Each $\alpha \in (0, 1)$ is attained by placing a fraction $\alpha_N \rightarrow \alpha$ of particles at $1 - \varepsilon_N$ and the remainder at $1 + \frac{\alpha_N}{1-\alpha_N}\varepsilon_N$, with mean exactly one, $\alpha_N N$ integral and $\varepsilon_N \downarrow 0$. In the absence of convergence of the boundary proportion, distinct subsequences may yield distinct values of α . Weight sequences converging to δ_1 in law therefore admit residual count limits ranging from the deterministic identity to the full multinomial count law: vanishing weight heterogeneity does not imply vanishing residual randomization.

V. THE LIMITING SPECTRAL DISTRIBUTION UNDER MULTINOMIAL RESAMPLING

Theorem 11 (Limiting spectral distribution; identity population). *Assume multinomial resampling, (A3)–(A4), $\Sigma_N = I_d$, and the entry convention of Section II with finite fourth moment. Then the empirical spectral law of $S_{r,N}$ converges in probability to $\mu_Q = \mathcal{T}_c(\nu)$ with $\nu = \text{MixPoi}(\rho)$.*

The supplementary material details the transfer of the classical theorem to the random, unbounded count diagonal: truncation to $\nu^{(M)}$, a subsequence-and-conditioning bridge for the random D_N , and removal of the truncation by rank control together with continuity of the fixed point.

Theorem 12 (General population spectrum; Gaussian case). *Assume multinomial resampling, (A3)–(A5), and a standard Gaussian matrix Z_N independent of the weights and of the auxiliary resampling randomness. Then the limiting spectral distribution of $S_{r,N} = \Sigma_N^{1/2}(\frac{1}{N}Z_N D_N Z_N^\top)\Sigma_N^{1/2}$ is*

$$\mu_S = H \boxtimes \mu_Q = H \boxtimes \mathcal{T}_c(\nu). \quad (13)$$

Gaussianity is used at exactly one point. Conditionally on D_N , which is independent of Z_N by exogeneity, the matrix $Q_N = \frac{1}{N}Z_N D_N Z_N^\top$ is orthogonally invariant and hence asymptotically free from the deterministic Σ_N . Unbounded counts are handled by truncation and rank approximation before passing to the limit. No universality claim beyond the Gaussian case is made.

Remark 13 (Scope and normalization). Theorem 11 applies verbatim to $S_{w,N}$, giving $\mu_{\text{pre}} = \mathcal{T}_c(\rho)$. As a normalization check, $\mathcal{T}_c(\delta_1) = \text{MP}(c)$, which corresponds to deterministic counts $n_i \equiv 1$ and not to multinomial resampling; uniform weights under multinomial resampling give $\rho = \delta_1$ but $\nu = \text{Poi}(1)$, and hence a nontrivial distortion. For $c > 1$, the bound $\text{rank}(S_{r,N}) \leq N < d$ forces a zero atom of mass at least $1 - 1/c$ in $\mathcal{T}_c(\nu)$, which preserves nonnegativity in the companion identity.

Proposition 14 (Zero-eigenvalue mass and rank loss [25]). *Under the assumptions of Theorem 11,*

$$\mu_Q(\{0\}) = \max\left(0, 1 - \frac{1-\nu(\{0\})}{c}\right), \quad (14)$$

and the finite-sample rank satisfies $\text{rank}(S_{r,N}) \leq \min(d, \#\{i : n_i > 0\})$, with equality almost surely for continuously distributed Z in the identity-population case. In the separable case of Theorem 12, $\mu_S(\{0\}) = \max\{H(\{0\}), \mu_Q(\{0\})\}$ by the maximum-atom rule for free multiplicative convolution.

Exact finite-sample rank loss and the zero mass of the limiting spectral distribution are distinct notions and are established separately. We note that $\text{rank}(\Sigma_N)/d \rightarrow 1 - H(\{0\})$ need not hold under (A5) alone.

Proposition 15 (Spectral moments and the exact increment). *Under the assumptions of Theorem 12,*

$$m_{2,\text{pre}} = m_{2,H} + c a_1^2 r_2, \quad m_{2,\text{post}} = m_{2,H} + c a_1^2 (1 + r_2), \quad (15)$$

so that the multinomial increment equals $c a_1^2$. The first moment is preserved in the same limiting sense, with $m_1 = a_1$ on both sides. The normalized empirical first and second moments converge in probability, and not merely in expectation.

Proposition 16 (Information content of the effective sample size). *The effective sample size satisfies $N/\text{ESS} \rightarrow r_2$, equivalently $\text{ESS}/N \rightarrow 1/r_2$. Given c and H , ESS therefore determines the multinomial limiting second spectral moment, but it does not determine the full limiting spectral distribution. It also yields an occupancy bound: for $K \sim \text{MixPoi}(\rho)$ we have $\mathbb{E}K = 1$ and $\mathbb{E}K^2 = 1 + r_2$, so that $1 = (\mathbb{E}K)^2 \leq \mathbb{E}K^2 \cdot \mathbb{P}(K > 0)$, giving*

$$p_{\text{multi}} \geq \frac{1}{1+r_2}. \quad (16)$$

If $r_2 = 1$ with mean 1, then necessarily $\rho = \delta_1$ and the multinomial count law is determined, with $\nu = \text{Poi}(1)$.

In general, however, ESS determines neither the occupancy structure, nor the zero mass, nor the full spectrum, nor the residual post-resampling second moment, the last of which requires $\mathbb{E}_\pi U$. Explicitly, the two weight laws

$$\rho^{(A)} = \frac{1}{2}\delta_{1/4} + \frac{1}{2}\delta_{7/4}, \quad \rho^{(B)} = \frac{9}{10}\delta_{3/4} + \frac{1}{10}\delta_{13/4} \quad (17)$$

share mean 1 and second moment 25/16, and hence share the value $\text{ESS} = 16N/25$ exactly at multiples of 10 by a deterministic ratio construction, yet have count zero masses 0.476287 and 0.429007 and count third moments 8.375 and 9.5. Since $m_3(\mathcal{T}_c(\nu)) = 1 + 3cb_2 + c^2b_3$ with equal b_2 and distinct b_3 , the two limiting spectra differ for every $c > 0$; at $c = 1$ the zero-eigenvalue masses are 0.476287 and 0.429007, since then $\mu_Q(\{0\}) = \nu(\{0\})$.

Corollary 17 (Asymptotic decrease of the effective dimension). *Under the assumptions of Proposition 15 with $a_1 > 0$ and $c > 0$, set $d_{\text{eff}}(M) := (\text{tr } M)^2 / \text{tr } M^2$. Then*

$$\begin{aligned} \frac{d_{\text{eff}}(S_{w,N})}{d} &\xrightarrow{\mathbb{P}} \frac{a_1^2}{m_{2,H} + c a_1^2 r_2}, \\ \frac{d_{\text{eff}}(S_{r,N})}{d} &\xrightarrow{\mathbb{P}} \frac{a_1^2}{m_{2,H} + c a_1^2 (1+r_2)}, \end{aligned} \quad (18)$$

with explicit gap $c a_1^4 / [(m_{2,H} + c a_1^2 r_2)(m_{2,H} + c a_1^2 (1+r_2))] > 0$.

At the finite-sample level we assert only that the unnormalized $\text{tr } S_{r,N}^2$ is nondecreasing in conditional expectation. The conditional expectation of the inverse participation ratio $\text{tr } M^2 / (\text{tr } M)^2$ can decrease strictly already for $d = N = 2$.

VI. RANK LOSS AND THE SPECTRAL-GAP TRANSITION

Let $p_s := 1 - \nu_s(\{0\}) \in (0, 1]$ denote the limiting surviving-parent fraction, so that $p_{\text{multi}} = 1 - \mathcal{L}_\rho(1)$ for multinomial resampling.

Theorem 18 (Rank-and-spectral-gap transition [26]). *Assume multinomial resampling, (A3)–(A4), $\Sigma_N = I_d$, and the entry convention of Section II with finite fourth moment. The active index set is independent of the particle matrix by exogeneity, so that conditionally on the counts the active columns remain independent and identically distributed with the original entry distribution. Let $M_N := \#\{i : n_i > 0\}$, so that $M_N/N \rightarrow p$ in probability. Then the following hold.*

- (i) Subcritical regime $c < p$: $\mathbb{P}(\text{rank}(S_{r,N}) = d) \rightarrow 1$, and for every $\varepsilon > 0$,

$$\mathbb{P}(\lambda_{\min}(S_{r,N}) \geq (\sqrt{p} - \sqrt{c})^2 - \varepsilon) \rightarrow 1. \quad (19)$$

- (ii) Supercritical regime $c > p$: $\mu_Q(\{0\}) = 1 - p/c$, and for every $\varepsilon > 0$,

$$\mathbb{P}(\lambda_{\min}^+(S_{r,N}) \geq (\sqrt{c} - \sqrt{p})^2 - \varepsilon) \rightarrow 1. \quad (20)$$

- (iii) Criticality $c = p$: $\mu_Q(\{0\}) = 0$ and $\inf \text{supp } \mu_Q = 0$, so that the limiting spectral support touches zero without carrying an atom there, and

$$\lambda_{\min}^+(S_{r,N}) \xrightarrow{\mathbb{P}} 0. \quad (21)$$

Two clarifications are in order. First, the bounds in (i) and (ii) constitute an explicit universal lower bound, a Marchenko–Pastur lower-edge benchmark depending only on p . The true lower edge depends on the full count law and may be strictly larger than this benchmark; it is characterized by Proposition 19, and it is strictly larger at every off-critical grid point examined in Section VIII. Second, for general Σ_N no uniform lower gap can hold under (A5) alone; a gap statement in the separable case requires in addition that $\lambda_{\min}(\Sigma_N) \geq a_- > 0$.

Proposition 19 (Spectral edges and support [27], [28]). *Under the assumptions of Theorem 11, define*

$$z(u) := -\frac{1}{u} + y \int \frac{t}{1+tu} \nu(dt), \quad y = \frac{1}{c}, \quad (22)$$

on real $u \neq 0$ with $-1/u \notin \text{supp } \nu$. The support-complement parametrization uses the real branches on which $z'(u) > 0$; regular edges of the nonzero part of μ_Q arise at boundary critical points of these branches and take the form $c z(u^)$. Not every real root of $z'(u) = 0$ corresponds to an edge, and the critical zero endpoint, which corresponds to $u \rightarrow \infty$ rather than to a finite critical point, is treated separately in Theorem 18(iii). As a consistency check, $\nu = \delta_1$ gives $c(1 \mp \sqrt{1/c})^2 = (1 \mp \sqrt{c})^2$.*

Since $\nu = \text{MixPoi}(\rho)$ charges every nonnegative integer whenever $\rho((0, \infty)) > 0$, the limiting spectrum is unbounded on the right:

$$\sup \text{supp } \mathcal{T}_c(\text{MixPoi}(\rho)) = \infty, \quad (23)$$

and consequently $\lambda_{\max}(S_{r,N}) \rightarrow \infty$ in probability.

The proof of (23) exhibits a positive limiting fraction of eigenvalues above any prescribed level; a single escaping outlier would not suffice. There is thus no finite global right edge, the lower and internal edges remain characterized as above, and bulk extent must be treated as a quantile notion. Whether the support can separate into several intervals for particular ρ is left open.

VII. RESIDUAL RESAMPLING

Theorem 20 (Residual spectral transform). *Under (A3), (A4R), identity population covariance, and the entry assumptions of Theorem 11, the empirical spectral law of $S_{r,N}$ under residual resampling converges in probability to*

$$\mu_Q^{\text{res}} = \mathcal{T}_c(\nu_{\text{res}}). \quad (24)$$

If instead the population covariance satisfies (A5) and Z_N is Gaussian, under the same exogeneity and independence assumptions, $\mu_S^{\text{res}} = H \boxtimes \mathcal{T}_c(\nu_{\text{res}})$.

The transfer of the empirical second-moment convergence to the residual setting is established separately in the supplementary material and is not inherited from the statement on the limiting spectral distribution.

Corollary 21 (Residual zero atom and threshold). *We have*

$$\begin{aligned} \nu_{\text{res}}(\{0\}) &= \mathbb{E}_\pi[\mathbf{1}_{\{K=0\}} e^{-U}], \\ p_{\text{res}}(\pi) &:= 1 - \nu_{\text{res}}(\{0\}). \end{aligned} \quad (25)$$

Under the identity-population and entry assumptions of Theorem 18, with (A4R) in place of (A4), all three regimes of that theorem hold with p_{res} in place of p . The noncritical case transfers through active-column domination and the critical case through the residual limiting spectral distribution and its fixed-point equation.

Proposition 22 (Residual second-moment cost). *Under (A4R) we have $\mathbb{E}[K + P_U \mid K, U] = K + U$ and $\text{Var}(K + P_U \mid K, U) = U$, whence $b_2(\nu_{\text{res}}) = r_2 + \mathbb{E}_\pi U$. Under the residual analogue of the assumptions of Proposition 15,*

$$m_{2,\text{res}} - m_{2,\text{pre}} = c a_1^2 \mathbb{E}_\pi U \leq c a_1^2, \quad (26)$$

with strict inequality if and only if $a_1 > 0$ and $\mathbb{E}_\pi U < 1$. The comparison is one of asymptotic normalized second spectral moments; no ordering for arbitrary spectral functionals is claimed.

The cost vanishes if and only if $a_1 = 0$ or $\mathbb{E}_\pi U = 0$; assuming $a_1 > 0$, if and only if $\mathbb{E}_\pi U = 0$, that is, if and only if π is supported on $\{U = 0\}$. This is a property of the limit law and not of every finite sample. For $a_1 > 0$, equality with the multinomial asymptotic cost occurs precisely at the integer-boundary limit $\pi = \delta_{(0,1)}$, since $\mathbb{E}_\pi(K + U) = 1$ together

with $\mathbb{E}_\pi U = 1$ forces $U = 1$ and $K = 0$ almost surely under π . Exact zero cost at finite N holds whenever the counts are deterministic, in particular when every $\lambda_i \in \mathbb{Z}_{\geq 0}$, and under the geometric degeneracy of Remark 5.

Corollary 23 (Asymptotic rank-preservation ordering). *For any joint limit π ,*

$$\nu_{\text{multi}}(\{0\}) - \nu_{\text{res}}(\{0\}) = \mathbb{E}_\pi[\mathbf{1}_{\{K \geq 1\}} e^{-(K+U)}] \geq 0, \quad (27)$$

so that $\mu_{\text{res}}(\{0\}) \leq \mu_{\text{multi}}(\{0\})$: residual resampling preserves asymptotically at least as much rank as multinomial resampling. The statement orders limiting occupancy and the zero mass of the limiting spectral distribution; it is not a sample-path rank comparison.

The count-level inequality is strict if and only if $\pi(K \geq 1) > 0$. At the spectral level, writing $q_{\text{multi}} := (1 - p_{\text{multi}}/c)_+$ and $h_0 := H(\{0\})$, we have $\mu_{\text{res}}(\{0\}) < \mu_{\text{multi}}(\{0\})$ if and only if $\pi(K \geq 1) > 0$ and $q_{\text{multi}} > h_0$; in the identity-population case $h_0 = 0$ and the condition reduces to $c > p_{\text{multi}}$. On the window $c \in (p_{\text{multi}}, p_{\text{res}})$, multinomial resampling already carries a macroscopic zero atom while residual resampling has full rank with probability tending to one.

Corollary 24 (Exact-uniform-weight contrast). *Let $w_i = 1/N$ for every N . Residual resampling then leaves the second-moment matrix unchanged at every finite N , so that in the identity-population setting its limiting spectral distribution is $\text{MP}(c)$, with zero resampling cost and $p_{\text{res}} = 1$; in a general Gaussian population model the unchanged matrix has limiting spectral distribution $H \boxtimes \text{MP}(c)$. Multinomial resampling produces $\nu = \text{Poi}(1)$, cost $c a_1^2$, and the transition at $p_{\text{multi}} = 1 - e^{-1}$. As classified in Theorem 10, sequences satisfying the corresponding joint-limit and moment assumptions realize every intermediate residual limit $(1 - \alpha)\delta_1 + \alpha \text{Poi}(1)$ with $\alpha \in [0, 1]$.*

Corollary 25 (Finite-sample Frobenius ordering). *Under (A1)–(A2), set $k_i = \lfloor \lambda_i \rfloor$, $u_i = \text{frac}(\lambda_i)$, $L = \sum_i k_i$ and $R = \sum_i u_i$. When $L, R > 0$,*

$$\begin{aligned} \text{Cov}_{\text{multi}} - \text{Cov}_{\text{res}} &= \text{diag}(k) - \frac{kk^\top}{L} \\ &\quad + \frac{LR}{N} \left(\frac{k}{L} - \frac{u}{R} \right) \left(\frac{k}{L} - \frac{u}{R} \right)^\top \succeq 0. \end{aligned} \quad (28)$$

The boundary cases are immediate: if $R = 0$ then $\text{Cov}_{\text{res}} = 0$, and if $L = 0$ the two covariances coincide. Hence, by the quadratic form of Proposition 4 with $G \circ G$ positive semidefinite,

$$0 \leq \mathbb{E}[\|S_{\text{res}} - S_w\|_F^2 \mid X, w] \leq \mathbb{E}[\|S_{\text{multi}} - S_w\|_F^2 \mid X, w]. \quad (29)$$

Corollary 25 is a count-covariance-level comparison in the error energy associated with the second spectral moment cost of this paper. No ordering for arbitrary convex spectral functionals is claimed, and the result is not presented as a general variance-comparison theory for resampling schemes.

VIII. NUMERICAL STUDY

The numerical study is organized by the logical layers of the theory rather than by experiment size: the count laws of Section IV are examined first, then the spectral map of Definition 8, and finally the consequences established in Sections V–VII. Every quantity reported below is compared against a closed-form prediction fixed prior to the corresponding simulation, so that each comparison admits a pass-or-fail reading rather than a qualitative one.

A. Design and validation protocol

For each experiment the target quantities, sample sizes, repetition counts and acceptance criteria were fixed in advance; the theoretical values are evaluated in closed form from the statements of Sections IV–VII and are not fitted. All runs use a single fixed seed, no reported result is selected from a larger pool of runs, and each figure addresses a single question. Within each repetition the schemes are compared on paired particle clouds, so that differences between schemes are not confounded with variation between clouds, and confidence intervals are computed across independent repetitions and reported as one standard error of the mean. Atoms are plotted as separate point masses and are never merged into a density. Since Theorem 10 depends on the manner in which the weights approach the integers, integer parts are stored as integers and λ_i is never reconstructed from a floating-point product Nw_i , and the endpoint construction uses $N = 2^m$ so that $1 - 1/N$ and $2 - 1/N$ are exactly representable.

Discrepancy metric. Agreement between a simulated spectrum and a predicted law is measured by a fixed-bandwidth Stieltjes discrepancy

$$E_h := |m_{\text{emp}}(x + ih) - m_{\text{th}}(x + ih)|, \quad (30)$$

evaluated on a stated grid and reported in both maximum and grid-averaged form, together with a sensitivity check at bandwidth $h/2$. This choice avoids the bandwidth arbitrariness of density estimation while remaining sensitive to atoms and to edges. All spectral predictions are computed from the fixed-point equation of Definition 8. Since several classically equivalent companion-side forms coincide at $\eta = \delta_1$ while governing a different object in the present setting, recovery of the Marchenko–Pastur law alone does not validate an implementation; the solver therefore passes a battery of unit tests before any experiment is run, each reported with its realized tolerance in the supplementary material (S.23).

B. The count-law-to-spectrum map

Design. The configuration uses an identity population, Gaussian entries, $N = 1000$, $d = 800$ so that $c = 0.8$, and $n_{\text{rep}} = 20$. Two weight profiles are considered: exactly uniform weights, and the two-point law $\rho = \frac{1}{2}\delta_{1/2} + \frac{1}{2}\delta_{3/2}$ realized by an exact ratio construction of mean exactly one. For each profile the empirical count probability mass function is compared against $\text{MixPoi}(\rho)$ and ν_{res} , and the simulated spectra of the pre-resampling, multinomial and residual matrices are compared against their $\mathcal{T}_c$ predictions, with atoms drawn separately.

Predictions. All targets follow in closed form. With $H = \delta_1$, $a_1 = 1$, $m_{2,H} = 1$ and $c = 0.8$, Propositions 15 and 22 give

$$\begin{aligned} m_2^{\text{pre}} &= 1 + c r_2, & m_2^{\text{multi}} &= 1 + c(1 + r_2), \\ m_2^{\text{res}} &= m_2^{\text{pre}} + c \mathbb{E}_\pi U, \end{aligned} \quad (31)$$

while Theorems 7 and 9 give the count zero masses $\mathcal{L}_\rho(1)$ and $\mathbb{E}_\pi[\mathbf{1}_{\{K=0\}}e^{-U}]$, and Proposition 14 converts each into a spectral atom. For the uniform profile $r_2 = 1$ and $\mathbb{E}_\pi U = 0$; for the two-point profile $r_2 = 5/4$ and $\mathbb{E}_\pi U = 1/2$, since both atoms are located at fractional part $1/2$.

Results. Every pre-registered quantity agrees to within 2σ , and Table I reports the full comparison. Two features of the comparison are not apparent from the tabulated values alone. First, under exactly uniform weights the residual row does not merely agree with the pre-resampling row to within the reported error: it reproduces that row exactly, since the realized matrix is the same, as required by Corollary 24 at every finite N and not only in the limit. Second, the Stieltjes discrepancies at $h = 0.05$ have grid means between 0.0019 and 0.0040 and maxima of at most 0.0302, localized in the left boundary layer; halving the bandwidth raises the maxima to at most 0.0428 without altering any conclusion, which identifies the residual misfit as a boundary-layer effect rather than a bulk one.

Under the two-point profile the three schemes separate as predicted, with second moments 2.000, 2.800 and 2.400. Under exactly uniform weights, multinomial resampling nevertheless produces a zero atom of mass 0.21. This distortion is not a small-sample artifact: it is the value predicted by $\mathcal{T}_c(\text{Poi}(1))$.

The same framework predicts the rank-and-spectral-gap transition at $c = p_s$ and the conservativeness of the Marchenko–Pastur benchmark edge, which costs a factor of approximately 1.3 in the gap it certifies; both are verified over a twelve-point sweep in c and are reported in the supplementary material (S.24).

C. Identical effective sample size with different null spaces

Proposition 16 states that ESS determines the limiting second spectral moment given c and H , but does not determine the spectrum. The present experiment quantifies the resulting gap, which is of practical relevance because ESS is the standard scalar diagnostic for resampling.

Design. The pair

$$\rho^{(A)} = \frac{1}{2}\delta_{1/4} + \frac{1}{2}\delta_{7/4}, \quad \rho^{(B)} = \frac{9}{10}\delta_{3/4} + \frac{1}{10}\delta_{13/4} \quad (32)$$

of Proposition 16 shares mean 1 and second moment $25/16$, and hence shares the value $\text{ESS} = 16N/25$ whenever N is a multiple of 10, while having distinct count third moments 8.375 and 9.5. The configuration uses multinomial resampling with $c = 0.8$, $N = 1000$ and $n_{\text{rep}} = 30$. Four quantities are reported: ESS and the predicted m_2 , which coincide by construction; the count zero masses; and the spectral atoms. The count third moments enter through $m_3 = 1 + 3cb_2 + c^2b_3$.

Results. ESS equals 640 on both sides, verified as an identity rather than estimated, and the second spectral moments are statistically indistinguishable, with $|\Delta| = 0.0162$ against a

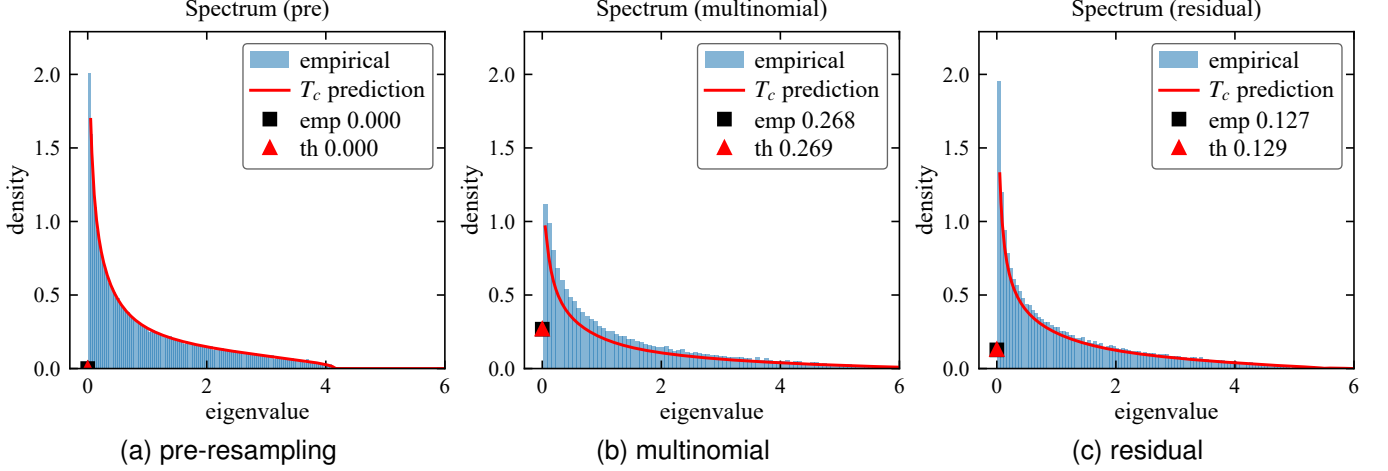


Fig. 1. One map, three count laws: simulated spectra against $\mathcal{T}_c$ predictions under the two-point weight law $\rho = \frac{1}{2}\delta_{1/2} + \frac{1}{2}\delta_{3/2}$, with an identity population and Gaussian entries ($N = 1000$, $d = 800$, $c = 0.8$, $n_{\text{rep}} = 20$). Zero atoms are drawn as separate point masses at the origin and are not merged into the density; all three panels share both axes. The three schemes separate as predicted, with second moments 2.000, 2.800 and 2.400. The exactly uniform profile is omitted, since there residual resampling reproduces the pre-resampling matrix itself and the two panels would coincide exactly, which is the content of Corollary 24; see Table I. The corresponding count probability mass functions are reported in the supplementary material (S.25).

TABLE I

EXPERIMENT E1: PRE-REGISTERED SPECTRAL QUANTITIES AGAINST THEORY. EMPIRICAL VALUES ARE MEANS OVER $n_{\text{rep}} = 20$ INDEPENDENT REPETITIONS, REPORTED WITH ONE STANDARD ERROR. HERE $\nu(\{0\})$ DENOTES THE COUNT ZERO MASS AND $\mu_Q(\{0\})$ THE INDUCED SPECTRAL ATOM. UNDER THE UNIFORM PROFILE THE RESIDUAL ROW REPRODUCES THE PRE-RESAMPLING ROW EXACTLY, AND NOT MERELY TO WITHIN THE REPORTED ERROR. THE FIRST SPECTRAL MOMENT IS 1.000 TO WITHIN MONTE CARLO ERROR IN EVERY ROW AND IS OMITTED.

Profile	Scheme	m_2		$\nu(\{0\})$		$\mu_Q(\{0\})$	
		empirical	theory	empirical	theory	empirical	theory
uniform	pre	1.8039 ± 0.0012	1.800	0	0	0	0
uniform	multinomial	2.6118 ± 0.0074	2.600	0.36805 ± 0.00195	0.367879	0.21006 ± 0.00244	0.209849
uniform	residual	1.8039 ± 0.0012	1.800	0	0	0	0
two-point	pre	2.0011 ± 0.0019	2.000	0	0	0	0
two-point	multinomial	2.7985 ± 0.0103	2.800	0.41215 ± 0.00230	0.414830	0.26519 ± 0.00288	0.268538
two-point	residual	2.4026 ± 0.0057	2.400	0.30580 ± 0.00148	0.303265	0.13225 ± 0.00184	0.129081

TABLE II

EXPERIMENT E3: IDENTICAL ESS WITH DIFFERENT NULL SPACES. EMPIRICAL VALUES ARE MEANS OVER $n_{\text{rep}} = 30$ REPETITIONS, REPORTED WITH ONE STANDARD ERROR. ESS IS VERIFIED AS AN EXACT IDENTITY AND IS NOT ESTIMATED.

Quantity	Profile A	Profile B
ESS (exact)	640	640
m_2 empirical	3.0559 ± 0.0101	3.0397 ± 0.0161
m_2 theory	3.05	3.05
$\nu(\{0\})$ empirical	0.47717 ± 0.00143	0.42823 ± 0.00215
$\nu(\{0\})$ theory	0.476287	0.429007
$\mu_Q(\{0\})$ empirical	0.34646 ± 0.00179	0.28529 ± 0.00268
$\mu_Q(\{0\})$ theory	0.345359	0.286259
m_3 empirical	12.576 ± 0.097	13.171 ± 0.180
m_3 theory	12.510	13.230

joint 2σ of 0.0381. The null spaces are not indistinguishable: the spectral atoms separate by 0.0612 against a predicted 0.0591, or nineteen standard errors, and the count third moments corroborate the mechanism. Table II reports the full

comparison.

Two weight configurations that any diagnostic based on ESS reports as identical therefore differ by six percentage points of the ambient dimension in the fraction of directions annihilated by a single resampling step. The quantity involved is finite and directly measurable rather than asymptotic.

D. Residual randomization at the integer boundary

Design. The three constructions of Theorem 10 are used, all satisfying $\rho_N \Rightarrow \delta_1$: exactly uniform weights, giving $\alpha = 0$; the interior family at $\alpha = 0.4$, in which a fraction $\alpha_N \rightarrow \alpha$ of particles is placed at $1 - \varepsilon_N$ and the remainder at $1 + \frac{\alpha_N}{1 - \alpha_N} \varepsilon_N$, so that

$$\frac{1}{N} \sum_i \lambda_i^2 = 1 + \frac{\alpha_N}{1 - \alpha_N} \varepsilon_N^2, \quad \frac{R_N}{N} = \alpha_N; \quad (33)$$

and the bounded two-level endpoint construction $\lambda_i = 1 - \frac{1}{N}$ for $i < N$ with $\lambda_N = 2 - \frac{1}{N}$ and $N = 2^m$, giving $\alpha = 1$. Each construction is run at $N = 2^8, \dots, 2^{14}$. The reported quantities are count-level and therefore inexpensive to evaluate: R_N/N , the zero-count fraction, and the second count moment, each

TABLE III

EXPERIMENT E4: COUNT-LEVEL QUANTITIES FOR THE THREE CONSTRUCTIONS OF THEOREM 10, AT THE ENDPOINTS OF THE SWEEP; THE FULL SWEEP $N = 2^8, \dots, 2^{14}$ IS TABULATED IN THE SUPPLEMENTARY MATERIAL (S.26). FOR THE UNIFORM CONSTRUCTION THE COUNTS ARE DETERMINISTIC, SO THAT ALL ENTRIES ARE EXACT AND CARRY NO MONTE CARLO ERROR.

Construction	N	R_N/N	zero fraction	$b_{2,N}$	$\frac{1}{N} \sum (\lambda_i - 1)^2$
uniform	all	0	0	1	0
	(theory)	0	0	1	—
interior	256	0.3984	0.15408 ± 0.00089	1.3963 ± 0.0037	2.59×10^{-3}
	16384	0.4000	0.14835 ± 0.00012	1.3998 ± 0.0005	4.07×10^{-5}
	(theory)	0.4	0.147152	1.4	$\rightarrow 0$
endpoint	256	0.9961	0.36859 ± 0.00135	2.0034 ± 0.0062	3.89×10^{-3}
	16384	0.9999	0.36800 ± 0.00018	2.0007 ± 0.0008	6.10×10^{-5}
	(theory)	1	0.367879	2	$\rightarrow 0$

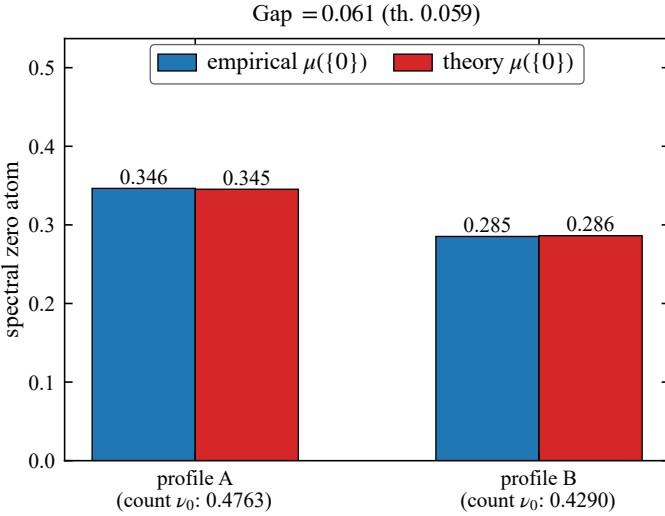


Fig. 2. Two weight profiles with exactly equal ESS, of value 640, and statistically indistinguishable second spectral moments, but macroscopically different null spaces, under multinomial resampling with $c = 0.8$, $N = 1000$ and $n_{\text{rep}} = 30$. The spectral atoms differ by 0.061 against a pre-registered prediction of 0.059, a separation of nineteen standard errors. Any diagnostic based on ESS reports the two configurations as identical.

compared against the predictions α , αe^{-1} and $1 + \alpha$ of Theorem 10 and Proposition 22, and displayed alongside the second-order weight deviation $\frac{1}{N} \sum_i (\lambda_i - 1)^2$.

Results. The second-order weight deviation vanishes at rate $1/N$ in all three constructions: for the interior family it decreases from 2.59×10^{-3} at $N = 256$ to 4.07×10^{-5} at $N = 16384$, a factor of 64 across a factor of 64 in N . Over the same range R_N/N converges to the three distinct limits 0, 0.400 and 1, and Table III reports the count-level quantities against their predictions α , αe^{-1} and $1 + \alpha$.

Weight heterogeneity, as measured by any moment of $\rho_N - \delta_1$, may therefore vanish while residual resampling retains the full randomization of the multinomial scheme. Since the endpoint construction keeps all weights below 2, the effect is governed by the position of the weights relative to the integer grid and not by heavy tails, and near-uniformity of the weights does not permit the inference that residual resampling is nearly deterministic.

E. The residual atom-free window

Design. A single spectral configuration of moderate size is used, with $N = 1024$, $d = 819$ so that $c = 0.7998$, and the interior construction at $\alpha = 0.5$. The pre-resampling, multinomial and residual spectra are compared against their $\mathcal{T}_c$ predictions. The value $\alpha = 0.5$ is used here in place of the value $\alpha = 0.4$ of Fig. 3 because it places c within the window of Corollary 23 with a wider margin; the two values yield the same qualitative conclusion.

Prediction. At $\alpha = 0.5$ the residual count law is $\frac{1}{2}\delta_1 + \frac{1}{2}\text{Poi}(1)$, so that

$$p_{\text{res}} = 1 - \frac{\alpha}{e} = 0.8161, \quad p_{\text{multi}} = 1 - e^{-1} = 0.6321. \quad (34)$$

With $c = 0.7998$ the configuration lies within the window $c \in (p_{\text{multi}}, p_{\text{res}})$ of Corollary 23. The multinomial spectrum must therefore carry an atom of mass $1 - p_{\text{multi}}/c = 0.210$, the residual spectrum must carry none, and the pre-resampling spectrum is $\text{MP}(c)$ without an atom, since $c < 1$.

Results. The three panels realize the prediction: the multinomial spectrum carries an atom of empirical mass 0.217 against the predicted 0.210, the residual spectrum carries none, and the pre-resampling spectrum recovers $\text{MP}(c)$. Within this window the two schemes differ qualitatively rather than quantitatively, since multinomial resampling has already eliminated a fixed fraction of directions while residual resampling retains full rank with probability tending to one. For $\alpha = 0.5$ the window spans $c \in (0.632, 0.816)$, a range that includes many practical ratios of particle number to dimension.

F. The finite-sample moment identity

Design. Sections V–VII are asymptotic, whereas Proposition 4 and Corollary 25 are not, and are therefore tested separately at a single fixed configuration in which no limit is taken. One fixed pair (X, w) with $d = 150$ and $N = 300$, for which the realized $R_N = 124$, is held constant while only the resampling randomness is redrawn, over $n_{\text{rep}} = 1000$ repetitions, for the multinomial and residual schemes. The residual count covariance is $\text{Cov}(\mathbf{n}) = \text{diag}(u) - uu^\top/R_N$, with the branch $R_N = 0$ defined to be zero. The cost is evaluated as

$$\text{tr } S_r^2 = \frac{1}{N^2} \mathbf{n}^\top (G \circ G) \mathbf{n}, \quad (35)$$

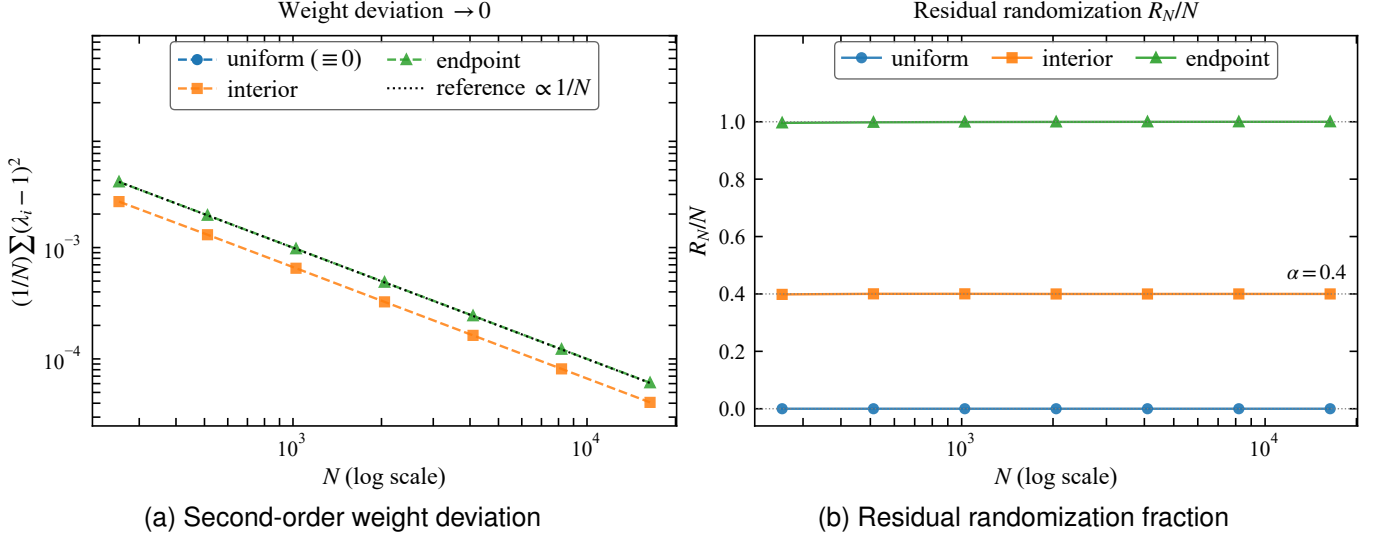


Fig. 3. Vanishing weight heterogeneity does not imply vanishing residual randomization, in accordance with Theorem 10. (a) The second-order deviation $\frac{1}{N} \sum_i (\lambda_i - 1)^2$ vanishes at rate $1/N$ in all three constructions, with the dotted line giving the reference slope. The uniform construction is omitted from this panel and marked as identically zero in the legend, since its deviation vanishes exactly at every N and cannot be displayed on a logarithmic axis. (b) Over the same range, R_N/N converges to the three distinct limits 0, 0.4 and 1. The two panels are to be read together: the simultaneous vanishing of weight heterogeneity and persistence of randomization constitutes the content of Theorem 10. Since the endpoint construction keeps all weights below 2, the effect originates in proximity to the integer grid rather than in heavy tails.

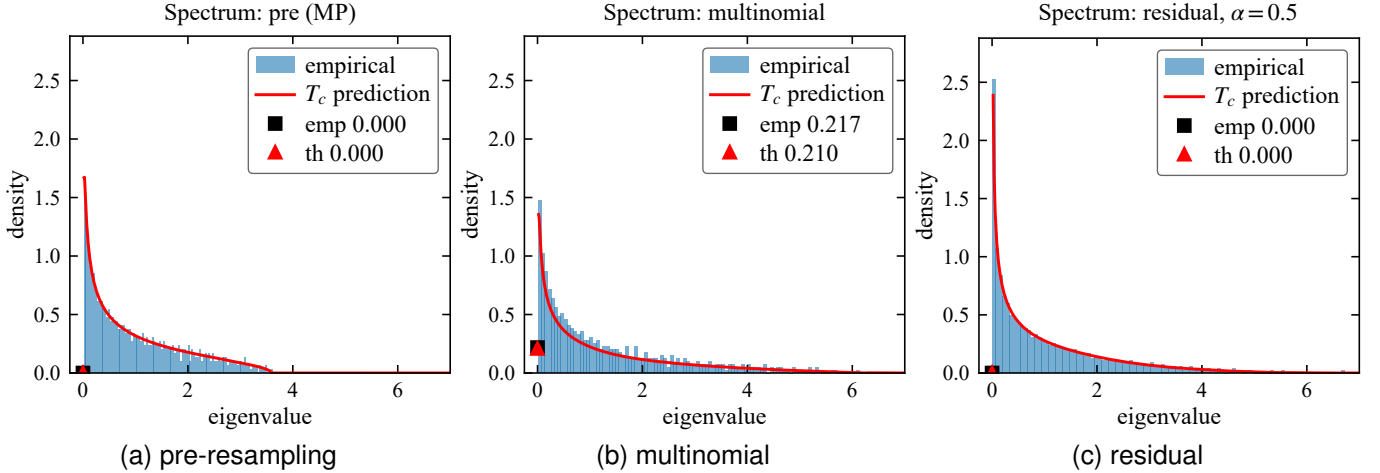


Fig. 4. The atom-free window of Corollary 23, at $N = 1024$, $d = 819$ and $c = 0.7998$, with the interior construction at $\alpha = 0.5$. The value of α differs from that of Fig. 3, where $\alpha = 0.4$; the larger value places c near the centre of the window $(p_{\text{multi}}, p_{\text{res}}) = (0.632, 0.816)$. All three panels share both axes. (a) The pre-resampling spectrum recovers MP(c). (b) Multinomial resampling carries a zero atom of empirical mass 0.217 against the predicted 0.210. (c) Residual resampling carries no atom: within this window it retains full rank with probability tending to one, whereas multinomial resampling has already annihilated a fixed fraction of directions.

which requires no eigendecomposition and therefore tests the identity in the form in which it is proved.

Results. At this configuration the standardized deviations are 1.948 for multinomial resampling, with Monte Carlo mean 74.114 ± 0.520 against a theoretical cost of 75.128, and 0.123 for residual resampling, with 31.299 ± 0.248 against 31.329; both lie within the pre-registered criterion $|\cdot| < 3$, so the residual covariance formula is confirmed in the exact form used in the proof. The covariance difference of Corollary 25 is positive semidefinite to machine precision, with minimum eigenvalue -2.7×10^{-15} , and since $G \circ G \succeq 0$ the finite-sample Frobenius ordering between the two schemes follows.

The statement is conditional and distribution-free, and requires none of the asymptotic assumptions.

G. Robustness checks

Three checks are summarized here and reported in full in the supplementary material (S.27). Repeating the E1 measurements at $N = 500$ and $N = 2000$ shows every quantity converging toward its prediction with diminishing error bars, the uniform-profile count zero mass moving from 0.3646 ± 0.0051 to 0.3701 ± 0.0020 against e^{-1} . Replacing Gaussian entries by variance-one Rademacher and Student- t_8 entries leaves the second spectral moments at 2.821 ± 0.017

and 2.821 ± 0.021 , against the finite- N predictions 2.798 and 2.801 of Proposition 6, which depend on the entry kurtosis κ_4 ; the deviations are 1.3σ and 0.9σ . This is an agreement report for a single configuration and is not offered as evidence of universality, and no universality claim is made in this paper. Finally, for $H = \frac{1}{2}\delta_{1/2} + \frac{1}{2}\delta_{3/2}$ the separable prediction $\mu_S = H \boxtimes \mathcal{T}_c(\text{Poi}(1))$ of Theorem 12 is verified end to end, with post-resampling $m_2 = 2.856$ against 2.85, a zero atom of 0.2079 against 0.209849, and the $\boxtimes$ pipeline matching a directly simulated Wishart spectrum to a maximum Stieltjes deviation of 5.1×10^{-4} .

IX. DISCUSSION AND LIMITATIONS

The results above concern four distinct objects, which should not be conflated [29]: the rank and zero-atom transition, described by $\mu(\{0\})$ with threshold p_s ; the universal Marchenko–Pastur lower-edge benchmark $|\sqrt{c} - \sqrt{p_s}|^2$; the regular lower and internal spectral edges, characterized by a branch-restricted critical equation; and two notions of conditioning, namely the raw pseudo-condition number $\kappa^+(S)$ and the bulk quantile condition number

$$\kappa_{\alpha,\beta}^{\text{bulk}} := Q_{\mu^+}(\beta)/Q_{\mu^+}(\alpha), \quad 0 < \alpha < \beta < 1, \quad (36)$$

where $\mu^+ := \mu|_{(0,\infty)}/(1 - \mu(\{0\}))$.

The scope of the two conditioning notions depends on the mechanism. For multinomial resampling in the identity-population model the limiting right support is unbounded by Proposition 19 and $\kappa^+(S_{r,N}) \rightarrow \infty$ in probability, since $\lambda_{\max}(S_{r,N}) \rightarrow \infty$ while $\lambda_{\min}^+(S_{r,N}) = O_{\mathbb{P}}(1)$. Fixed-quantile ratios therefore describe bulk conditioning more robustly than the raw pseudo-condition number, and a positive lower gap does not imply asymptotic well-conditioning. Residual limits may instead have bounded support, as in the exact-uniform-weight case, where κ^+ has a finite limit for $c \neq 1$. The spectral cost considered in this paper refers to these specific observables and not to generic performance metrics: given c and H , ESS determines the multinomial limiting second spectral moment and bounds occupancy from below, but does not determine the full limiting spectral distribution.

Five limitations delimit the scope of the results.

First, the analysis treats a single resampling step with exogenous weights. Endogenous, likelihood-driven weights generally invalidate the independence assumptions used in the covariance-spectrum step, so the spectral-map argument does not extend automatically to that setting. Conditionally on the realized weights the multinomial sampling law is unchanged, so suitable count-law limits may survive under corresponding random-weight assumptions.

Second, the limit theorems cover multinomial resampling under (A4) and residual resampling under (A4R). Establishing count-law limits for stratified and systematic resampling requires additional analysis of ordering and dependence and lies outside the scope of this paper; a random-ordering assumption may prove useful, but is not claimed to be sufficient. These two schemes accordingly do not appear in the numerical study.

Third, centering is not performed. It perturbs each matrix by a term of rank at most one and therefore does not affect

the limiting spectral distribution as $d \rightarrow \infty$ whenever that distribution exists, but the finite-sample conditional inequalities are sensitive to it.

Fourth, Theorem 12 is stated under Gaussianity alone, and no universality claim is made; the entry-distribution check of Section VIII-G covers a single configuration and is not a universality result.

Fifth, the results establish no ordering of filtering accuracy, closed-loop performance, or optimal resampling schedules: the experiments verify the objects that the theory defines and no others. The critical-point readings in Fig. 4 are finite- N measurements taken near a transition, and are reported together with their finite-size band rather than smoothed. The spectral recursion of a complete filtering cycle remains for future work.

X. CONCLUSION

Resampling mechanisms admit a classification by their count-law limits. Multinomial resampling is determined by the weight law ρ alone, whereas residual resampling retains information about the approach to the integer boundaries through the joint law π , classified completely under $\rho = \delta_1$ as the family $(1 - \alpha)\delta_1 + \alpha \text{Poi}(1)$. Vanishing weight heterogeneity therefore does not imply vanishing residual randomization.

Composed with a classical covariance-side spectral map, these count laws yield exact spectral consequences. For multinomial resampling in the identity-population model they yield a rank-and-spectral-gap transition with critical support touching, computable regular spectral edges, unbounded right support with diverging pseudo-condition number, and an exact second-moment cost. For residual resampling they yield the smaller cost $c a_1^2 \mathbb{E}_{\pi} U$, an asymptotic rank-preservation ordering, and a complete one-parameter classification of the limits under vanishing heterogeneity. Numerical experiments against pre-registered predictions confirm each level of the theory.

These results support the use of spectral diagnostics alongside ESS in the assessment of resampling for high-dimensional ensembles. Their incorporation into adaptive resampling rules for likelihood-driven particle filters remains an important direction for future work.

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